

Color Associations in Abstract Semantic Domains

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Abstract

The embodied cognition paradigm has stimulated ongoing debate about whether sensory data – including color – contributes to the semantic structure of abstract concepts. Recent uses of linguistic data in the study of embodied cognition have been focused on textual corpora, which largely precludes the direct analysis of sensory information. Here, we develop an automated approach to multimodal content analysis that detects associations between words based on the color distributions of their Google Image search results. Crucially, we measure color using a transformation of colorspace that closely resembles human color perception. We find that words in the abstract domains of academic disciplines, emotions, and music genres, cluster in a statistically significant fashion according to their color distributions. Furthermore, we use the lexical ontology WordNet and crowdsourced human judgments to show that this clustering reflects non-arbitrary semantic structure, consistent with metaphor-based accounts of embodied cognition. In particular, we find that images corresponding to more abstract words exhibit higher variability in colorspace, and semantically similar words have more similar color distributions. Strikingly, we show that color associations often reflect shared affective dimensions between abstract domains, thus revealing patterns of *aesthetic coherence* in everyday language. We argue that these findings provide a novel way to synthesize metaphor-based and affect-based accounts of embodied semantics.

1. Introduction

Color has been harnessed as a means of communication throughout human history, as evidenced by its use in painting, poetry, fashion, architecture, and marketing (Lakoff & Turner, 1989; Riley, 1995; Labrecque & Milne, 2012). Indeed, recent work shows that color can prime a range of attentional, emotional, and interpretive responses (Hill & Barton, 2005; Mehta & Zhu, 2009; Labrecque & Milne, 2012; Elliot & Maier, 2014). Color is also frequently used in linguistic metaphors to describe concepts across varying levels of abstraction, including those that lack direct visible referents (e.g., “I could play the blues and then not be blue anymore” — B.B. King) (Lakoff & Turner, 1989; Winter et al., 2018; Winter, 2019). A number of studies have found that linguistic metaphors can activate neurocognitive processes similar to those involved in forms of synaesthesia that ascribe colors to letters or numbers (Marks, 1982; O’Dowd et al., 2019). These findings are consistent with theories of multimodal cognition which argue that the metaphorical use of sensory data in everyday language indicates that sensory data plays a key role in the semantics of both concrete and abstract concepts (Tanenhaus et al., 1995; Barsalou, 2003, 2010; Gallese & Lakoff, 2005; Bergen, 2012; Dancygier & Sweetser, 2012).

However, popular theories maintain that the content and structure of concrete and abstract concepts are qualitatively different (Wiemer-Hastings & Xu, 2005; Binder et al., 2005; Brysbaert et al., 2014). Concepts are said to be concrete if their meanings involve perceivable features and referents (e.g. the concept “dog”), whereas abstract concepts are said to lack specific referents in sensory experience and to convey meanings that depend on theoretical knowledge (e.g. the concept “democracy”). A number of theories hold that abstract concepts are defined by formal, logical relations whose structure is independent of sensory data (Newell et al., 1989; Jackendoff, 2002; Chatterjee, 2010). For example, nativists frequently argue that abstract concepts are defined *a priori* by an internal formal language (Fodor, 1975; Laurence & Margolis, 2002; Medin & Atran, 2004), and connectionists view abstract concepts as amodal graph-theoretic objects (Tenenbaum et al., 2011). Recent work attempts to reconcile these views in proposing that while abstract concepts may not be represented using sensory data, they may be embodied through associations with emotion (Vigliocco et al., 2009; Kousta et al., 2011; Troche et al., 2014, 2017). Thus, there is ongoing debate about whether color (and sensory data in general) contributes to semantic associations between abstract concepts.

The recent emergence of ‘multimodal’ approaches in natural language processing (NLP) – i.e. approaches that integrate both text and images – hold promise in advancing this inquiry. These developments mark a critical advance upon largely text-based approaches, such as the qualitative identification of verbal metaphors (Lakoff & Turner, 1989; Gallese & Lakoff, 2005), and the quantitative measure of co-occurrence patterns among words in text using LDA (Blei & Ng, 2003) or word2vec (Mikolov et al., 2013). New models of *multimodal distributional semantics* show how images can be used to infer semantic associations among words (Bruni et al., 2014); however, these methods exploit a large number of (often uninterpretable) features to associate words with images (Deng et al., 2010; Lindner et al., 2012; Bruni et al., 2014). Moreover, these approaches are not explicitly designed to encode information that is relevant for human semantic processing. For example, they often operate in RGB colorspace, which does not accurately represent human color perception (International Commission on Illumination 1978; Safdar et al., 2017).

Here, we develop a computational method for quantifying statistical relationships between words and their color distributions, which we measure using a state-of-the-art transformation of colorspace that accurately captures human color perception (referred to as a *perceptually uniform* colorspace) (Safdar et al., 2017). We infer hierarchical clustering among words according to the perceptually uniform color distributions of their Google Image search results. By examining patterns of image production and search mediated by human users, and by representing color in a perceptually grounded fashion, our analyses are interpretable with respect to cognition (Lupyan & Goldstone, 2019).

For a range of concrete and abstract domains, our method extracts taxonomies from the lexical database WordNet and uses the words in these taxonomies as search terms to automatically

collect images from Google. We then associate each term with measures of semantic abstractness using WordNet’s hierarchical structure (Iliev & Axelrod, 2017) and judgments of concreteness from human coders (Brysbaert et al., 2014). This allows us to test whether the abstractness of a concept is inversely related to the amount of sensory information that it encodes (Barsalou, 2003; Wiemer-Hastings & Xu, 2005; Binder et al., 2005). To date, this hypothesis has been stated only in general terms with respect to concept imageability (Vigliocco et al., 2009; Kousta et al., 2011; Troche et al., 2014, 2017). To our knowledge, we are the first to test this hypothesis in a specific sensory domain (i.e. color), where we predict that increasing word abstractness increases the color variability of associated online images. We test this using the concrete domain of *animals* and the abstract domains of *academic disciplines*, *emotions*, and *music genres*.

Importantly, theories of embodied cognition provide compelling evidence that sensory data participates in abstract representation in non-arbitrary ways, as reflected by its use in popular linguistic metaphors (Lakoff & Turner, 1989; Gallese & Lakoff, 2005; Winter, 2019). Building on this intuition, we test not only whether color is used to represent abstract concepts, but also whether it encodes abstract relations between concepts, such as hierarchical information and semantic similarity. To do this, we measure whether each abstract domain is organized into distinct clusters of concepts based on color, and we determine whether these clusters are semantically coherent by testing if the proximity of words in colorspace correlates with text-based measures of their semantic similarity. We then examine whether color plays a metaphorical role in associating concepts from different abstract domains.

Across all domains examined, we find that color variability increases with semantic abstractness in a manner that correlates with crowdsourced human judgments. Nevertheless, consistent with theories of embodied cognition, we find that color associations in online images reflect coherent semantic relations among abstract concepts. We show that words cluster by color in a highly non-random fashion within each of the abstract domains examined, and that semantic similarity predicts color similarity. Finally, we examine color associations between abstract domains and identify patterns of *aesthetic coherence*, which we define as the use of color to reflect both common conceptual structure and shared affective dimensions between domains.

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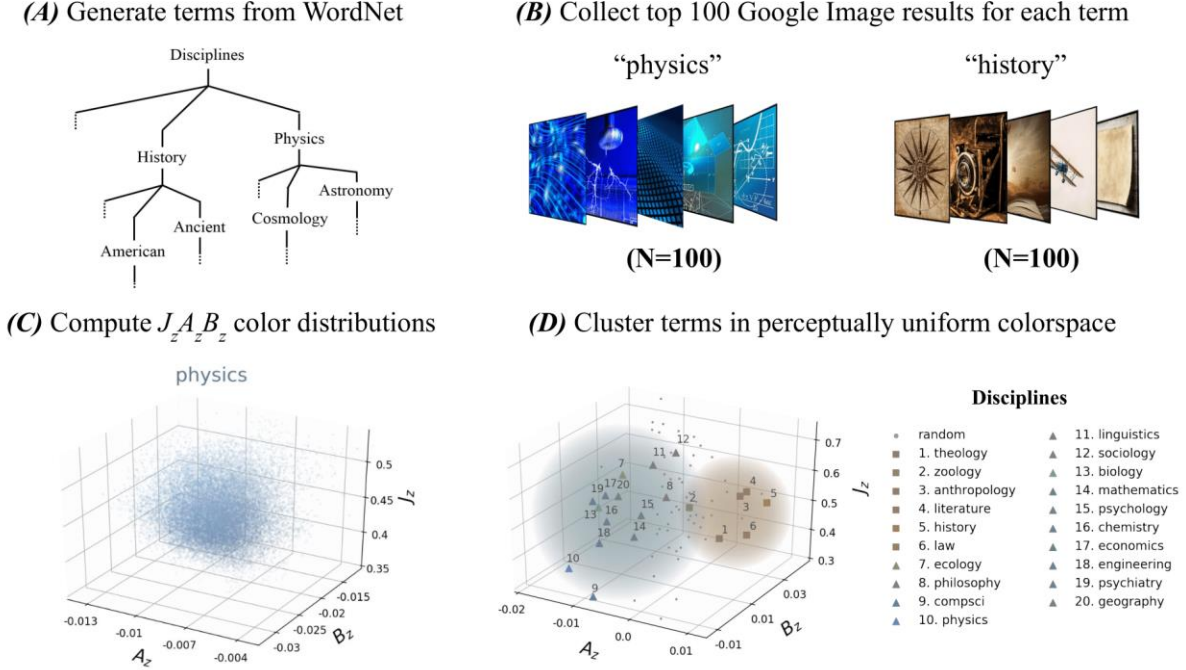


Fig. 1. Visualization of our analysis pipeline. (A) For each domain (e.g., *disciplines*), we generate search terms from the lexical database WordNet. (B) For each term within a domain (e.g., “physics”), we collect the top 100 Google Image search results. (C) Next, we transform each sRGB color distribution into the perceptually uniform $J_z A_z B_z$ colorspace. (D) We then average the color distributions over images for each search term to obtain aggregate perceptually uniform color distributions, and we compute the Jensen-Shannon divergence between the color distributions for pairs of terms to compare and cluster words in perceptually uniform colorspace. In panel (D), we visualize the domain of academic disciplines using the mean $J_z A_z B_z$ coordinates among Google Image search results for each word. Points are colored according to their mean $J_z A_z B_z$ coordinate. Spheres indicate clusters returned by our hierarchical analysis. Small markers indicate a sample of random WordNet nouns.

2. Materials and Methods

2.1. Overview. For each domain (e.g. *disciplines*), we extract search terms from WordNet. Then, we collect the top 100 Google Image search results for each term in each domain (e.g. “physics”, “jazz”, “love”). We compute the aggregate perceptually uniform color distributions for each term by averaging over its images, and we compare the color distributions of terms in each abstract domain. We cluster terms in each domain using the Jensen–Shannon (JS) divergence between their color distributions, and we compare JS divergence to measures of semantic similarity. We

now describe each step in detail (Fig. 1). Our analyses can be reproduced and extended using the Python package *comp-syn* (“Computational-Synesthesia”)¹.

2.2. Semantic Domain and Search Term Construction. We identify search terms for each domain by retrieving the hyponyms from WordNet corresponding to *disciplines*, *emotions*, and *music genres* (e.g. “classical” for *music genres*). We associate each term with an abstractness score based on its hierarchical position in WordNet (2.5. *Domain Analysis*). We compare results from these abstract taxonomies to a concrete domain by examining a sample of four subdomains of the *animal* hierarchy in WordNet – i.e., *dogs*, *cats*, *horses*, and *marsupials*. The domain of *animals* was selected for this analysis because, *prima facie*, it provides a representative example of a widely familiar concrete domain. These abstract domains were chosen because of their treatment in recent studies of concept concreteness (Brysbaert et al., 2014; Troche et al., 2017).

2.3. Google Data Acquisition. We use Google’s Python API to obtain the first 100 images returned by a Google search for each word. The order of Google search results is shaped by the PageRank algorithm, which incorporates search activity to identify which images users interact with most frequently (Jing & Baluja, 2008). All images were collected from the same U.S. IP address. The images associated with *disciplines*, *emotions*, and *music genres* were collected between June 18th and July 18th, 2019, and the images associated with the animal taxonomies were collected between November 20th and 28th. Our results are robust to the exclusion of repeat images and images that include branding (Fig. S3).

2.4. Computing Color Distributions. We use the PIL Python module to convert each image into an $m \times n \times 3$ array of sRGB values, where m and n are the intrinsic image dimensions. For efficiency, we compress each image into an anti-aliased $300 \times 300 \times 3$ array.

For each image, we transform each sRGB pixel value into its counterpart in perceptually uniform colorspace. We measure the density of points per image in eight evenly segmented regions of $J_zA_zB_z$ colorspace which are equally distinguishable by the human eye (Safdar et al., 2017). Finally, we compute the aggregate color distribution corresponding to each search term by averaging the $J_zA_zB_z$ distributions of its top 100 Google Image search results. For example, the $J_zA_zB_z$ distribution for “golden retriever” suggests that its images include more yellow than red and blue hues.

Our use of $J_zA_zB_z$ colorspace rather than RGB greatly increases the semantic coherence signal we report (Fig. S1,S2). This suggests our results are not driven by Google’s algorithm, because the image search algorithm leverages color in combination with many spatial features, and recent papers indicate that Google represents color using RGB (Jing & Baluja, 2008). The details of our compression, binning, and averaging steps do not affect our results (Fig. S3–S6).

¹ <https://github.com/comp-syn/comp-syn>

2.5. Domain Analysis. We use JS divergence to compare color distributions. For color distributions C_i and C_j associated with words i and j , the JS divergence is given by,

$$D_{JS}(C_1 || C_2) \equiv \frac{1}{2} [D_{KL}(C_1 || C_{12}) + D_{KL}(C_2 || C_{12})] \quad [1]$$

where D_{KL} is the Kullback-Leibler divergence and we have defined the mean color distribution $C_{12} \equiv (C_1 + C_2)/2$. JS divergence captures the distance between two color distributions, with lower values corresponding to more similar distributions in perceptually uniform colorspace.

We compare similarities among color distributions to the Jiang-Conrath measure of semantic similarity, which combines WordNet distances with corpus-based co-occurrence frequencies (Jiang & Conrath, 1997). We also adopt a popular measure of semantic abstractness called ‘branching factor’, computed from WordNet (Iliev & Axelrod, 2017). Branching factor assumes that supersets generalize over their subsets, making them more abstract (e.g. the abstract term “mammal” indexes a diverse set of animals that includes the much more specific subset of “chihuahuas”). Branching factor is the number of subsets indexed by a word,

$$B \equiv N_{descendants} + 1 \quad [2]$$

where the number of descendants $N_{descendants}$ is derived from WordNet. We benchmark this measure against crowdsourced judgments of concept concreteness from human coders; 78 categories (40 from concrete and 38 from abstract domains) could be mapped to concreteness ratings in Brysbaert et al. (2014).

2.6. Inferring Color Clustering. We test whether terms cluster in colorspace via the following steps: (i) for each taxonomy, compute the JS divergence between the color distributions of each pair of search terms; (ii) infer hierarchical clustering among search terms based on these JS divergence values using complete Euclidean agglomerative clustering (Everitt et al., 2011); (iii) compute the modularity of the hierarchical clustering partition produced in (ii) (Clauset et al., 2004). Our results are robust to alternative measures of clustering (*SI Appendix*).

To measure significance, we calculate the likelihood of obtaining the modularity observed for each taxonomy when sampling from a distribution of modularity values computed using the color distributions corresponding to 1000 subsamples of a set of random nouns from WordNet (*SI Appendix*). Our results hold under a wide range of prespecified cluster counts (*SI Appendix*). For convenience, we calculate modularities C_i using min-max normalization.

3. Results

Color variability in online images increases with the semantic abstractness of search terms. Yet, in abstract domains, search terms cluster by color at high statistical significance. The semantic similarity of search terms predicts their color similarity.

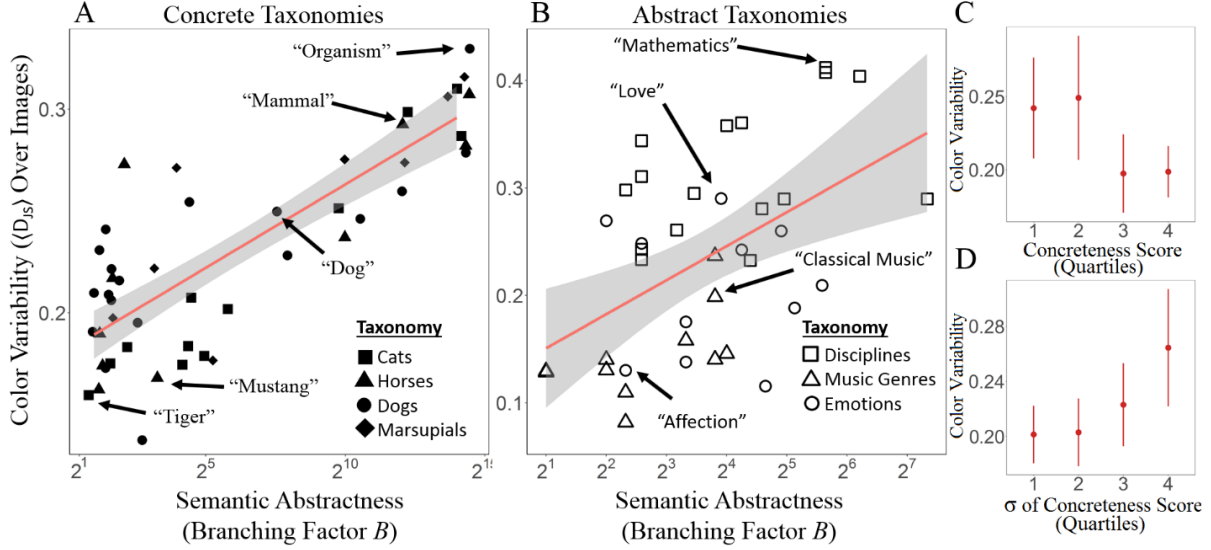


Figure 2: Average variability among the color distributions for Google Image search results for each search term versus the semantic abstractness of these terms. Color variability is measured as the mean JS divergence between the perceptually uniform color distributions of the top 100 Google Image results returned for each search term. Semantic abstractness is measured via WordNet branching factor, B . (A) The correlation between color variability and semantic abstractness for search terms within Wordnet’s lexical hierarchy for *animals*. (B) The correlation between color variability and semantic abstractness for search terms within the abstract taxonomy of disciplines, emotions, and music genres. (C) The correlation between crowdsourced concreteness ratings for search terms and their color variability, using both concrete and abstract taxonomies. (D) The correlation between the standard deviation of human concreteness ratings for search terms and their color variability, using both concrete and abstract taxonomies. Terminal categories with $B = 1$ are excluded from (A) and (B) due to heteroskedasticity that obscures the striking similarity in correlation between concrete and abstract domains; supplementary analyses indicate that these results are robust to including terminal categories.

3.1. Color variability increases with abstractness. We find a strikingly similar correlation between color variability and abstractness within the concrete domain of animals (Fig. 2A, $r_s=0.47$, $p<0.01$) and the abstract domains of disciplines, music genres, and emotions (Fig. 2B, $r_s=0.47$, $p<0.01$). These results hold when adjusting for outliers and heteroskedasticity from terminal categories with $B=1$ (Fig. S7,S8). Moreover, we find that human judgments of concept concreteness similarly predict color variation. Fig. 2C shows that search terms rated by people as more concrete are associated with lower color variability ($p<0.01$), and Fig. 2D shows that terms associated with greater variability in concreteness judgements are associated with greater color variability ($p<0.01$) (Wilcoxon Rank Sum).

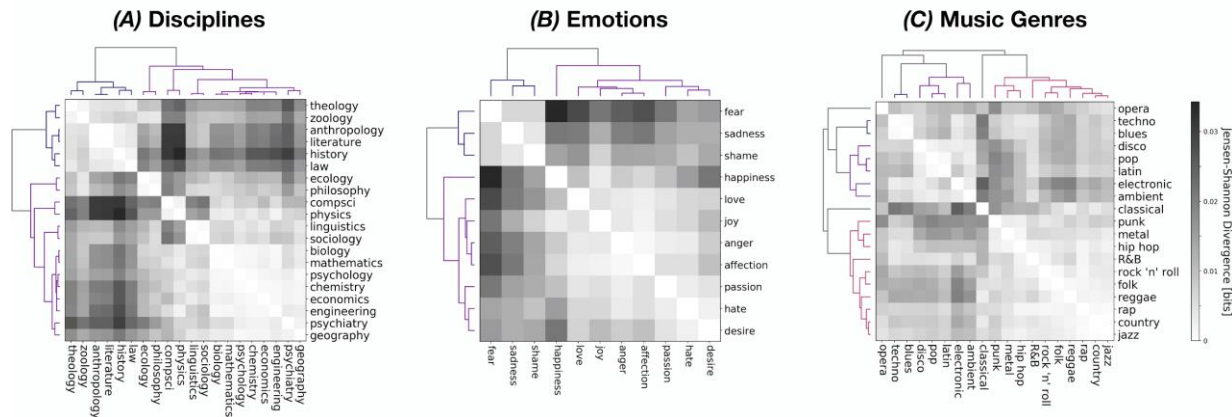


Fig. 3. Results of our color clustering analysis for (A) academic disciplines, (B) emotions, and (C) music genres. In each matrix, the colormap indicates the JS divergence (Eq. 1) between aggregate perceptually uniform color distributions for pairs of search terms. Low JS divergence values indicate similar color distributions; for example, Google Image search results for “Physics” and “Computer science” have very similar aggregate color distributions, while the distributions for “Happiness” and “Fear” are very different. Dendrograms indicate groups of like terms in colorspace determined by our hierarchical clustering analysis. Terms in each abstract domain cluster according to their perceptually uniform color distributions at high statistical significance relative to sets of random nouns.

3.2. Color Clustering. One may expect that greater color variability should prohibit coherent color associations among abstract concepts, but we find non-random clustering among abstract concepts in colorspace (Fig. 3). Inspection of our Google Image data suggests that this clustering signal is shaped by aesthetic choices in the design of images (Fig. S9).

For disciplines, we find two clusters distinguishing humanities and sciences. Humanities disciplines, including “History”, “Law”, and “Literature”, are associated with ‘warmer’ color distributions with reddish and brownish hues, while disciplines including “Chemistry”, “Physics”, and “Mathematics” exhibit ‘cooler’ color distributions with blueish and grayish hues (Fig. 1B). For emotions, we find three clusters: a cluster of blackish and reddish words including “Desire”, “Hate”, and “Passion”; a cluster of brighter reddish and yellowish words including “Happiness”, “Joy”, and “Love”; and a cluster of blueish and grayish words including “Fear”, “Sadness”, and “Shame”. Finally, we identify two music genre clusters: one with warmer hues including “Country”, “Folk”, and “Rock’n’roll”, and one with darker blueish hues including “Pop”, “Techno”, and “Blues”.

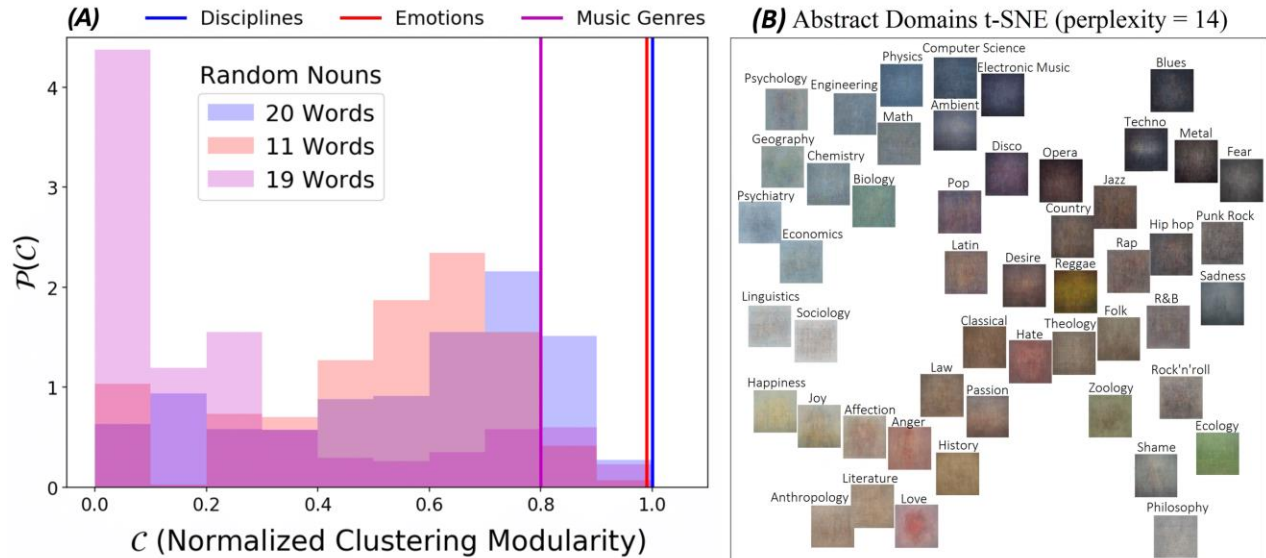


Fig. 4. (A) Color clustering results. Vertical lines show the normalized clustering modularities (Eq. 3) for the taxonomies of disciplines (blue), emotions (red), and music genres (magenta), and histograms show clustering modularity distributions for random WordNet nouns, where the number of random words corresponds to the size of each taxonomy. (B) t-SNE 2D projection of the eight-dimensional JZAZBZ color distribution for each search term in each abstract domain. Perplexity is set to 14. Proximity in the 2D projection indicates proximity in perceptually uniform colorspace. Each search term is represented by its colorgram, which refers to a composite image produced by averaging the color value for each pixel across all images in a search term’s image set.

Fig. 4A shows that clustering in each of the abstract taxonomies is significantly higher than expected based on a benchmark distribution of modularity values computed from random sets of WordNet nouns. For disciplines, the modularity of two clusters ($Q=0.04$, $N=20$) is higher than random ($p<0.01$); for emotions, the modularity of three clusters ($Q=0.05$, $N=11$) is higher than random ($p<0.01$); and for music genres, the modularity of two clusters ($Q=0.05$, $N=19$) is higher than random ($p<0.01$), (Centered Wilcoxon Signed Rank). These results are robust to parametric statistical tests (SI Appendix). Dimensionality reduction with t-SNE (Maaten and Hinton, 2008) yields qualitatively similar clustering patterns (Fig. 4B, Fig. S9).

3.3. Color Similarity Correlates with Semantic Similarity. The semantic coherence of our clustering results is corroborated by a significant correlation between color similarity and semantic similarity among search terms (Fig. S10, $r_s=0.43$, $p<0.01$).

4. Discussion

In this study, we find not only that abstract concepts are associated with colors in online images, but also that these color associations can encode relationships that have been traditionally viewed as symbolic and amodal in nature, such as hierarchical structure and semantic similarity. These

results provide novel evidence for the embodied view of cognition by suggesting that even abstract properties of semantic architecture, such as set-theoretic relations in lexical hierarchies, can participate in a multimodal interface with sensory information. As such, these findings reveal a new way to synthesize extant accounts of embodied cognition by suggesting that color can encode both metaphorical mappings (Gallese & Lakoff, 2005) and affective relations (Kousta et al., 2011; Troche et al., 2014, 2017) simultaneously. Here we discuss the ability for our method to identify *aesthetic coherence* in language, which we define as the use of sensory data to encode both logical and affective dimensions of semantic domains.

Patterns of aesthetic coherence are evident in the t-SNE visualization of abstract concepts according to their color distributions, displayed in fig. 4B. For example, in the domain of disciplines, emotions associated with reddish hues – “Love”, “Anger”, “Affection”, and “Passion” – cluster with qualitative disciplines, including “Literature”, “Anthropology”, “History”, and “Law.” By contrast, quantitative disciplines associated with bluish hues do not cluster with any of the emotion terms examined, consistent with empirical research indicating that bluish hues prime competence and rationality, rather than being highly emotive (Alberts & Geest, 2011; Labrecque & Milne, 2012; Mehta & Zhu, 2009). These results indicate that color can partition subclasses within abstract domains (e.g. qualitative and quantitative disciplines), while also reflecting affective contrasts among subclasses. Thus, while color associations among concrete concepts are likely mediated by intrinsic properties of objects, we find that color associations among abstract concepts can be mediated by non-arbitrary affective dimensions.

We pose a similar question regarding music: are color associations among music genres correlated with color associations among the emotions evoked by these genres? Excitingly, we find that genres which conventionally evoke aggressive and ‘dark’ themes (e.g. “Metal” and “Punk Rock”) cluster near the emotion “Fear” (fig. 4B). Meanwhile, genres which conventionally evoke romantic themes (e.g. “Latin”) cluster near the emotion “Desire” (Zentner et al., 2008).

Investigating the extent to which representations encoded by online search engines reflect representations at the individual level is a promising area for future study. Future experiments on the priming effects of abstract color associations are well-positioned to inform several unsolved puzzles (Mohammad, 2011), such as the finding that abstract concepts are cognitively retrieved more quickly than concrete concepts (Kousta et al., 2011; Vigliocco et al., 2009). Given that emotional valence has been proposed as an explanation of this finding (Kousta et al., 2011), we posit that aesthetic coherence may provide an important piece of the theoretical framework needed to understand such results more broadly.

Cultural biases may also contribute to the color associations we report. Notably, color associations among academic disciplines are correlated with cultural biases in gender representation and expression among the disciplines examined (Frassanito & Pettorini, 2008). Future work that applies our methods across cultures will provide insights into the potential universality of aesthetic coherence (Malt, 1995; Chloe et al., 2013; Youn et al., 2016).

Data availability: all datasets and code used in this analysis are publicly available at <https://github.com/bakerwho/comp-syn>.

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